

TRAINER NOTES

BEGINNER → ADVANCED

Microsoft Power BI

Complete Course Notes — Concepts, Interface, DAX, Visuals, Charts & Advanced Features

Based on official Microsoft Learn documentation ("Get started building with Power BI")
plus expanded trainer explanations, chart-usage guide, component reference and memory maps.

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1. What is Power BI? (The Big Picture)

Microsoft Power BI is a complete business-analytics (reporting) solution. Think of it as one toolbox with three jobs:

1. **Prepare data** — connect to messy, scattered data and clean it up.
2. **Visualize data** — turn numbers into charts, tables and dashboards that tell a story.
3. **Share & manage** — publish that story so the right people can see it, on the web or on a phone.

Trainer's plain-English definition: Power BI = Excel's reporting muscles + a website to publish reports + a mobile app to view them — all connected together.

Power BI is built to **scale**:

Use case	Example
Very simple	One person imports a single Excel sheet and builds a quick bar chart.
Very complex	An entire company connects 20+ databases, builds a governed data model with strict security, and distributes branded reports to thousands of employees company-wide.

Why it matters: whether you are a **Data Analyst**, **Business Analyst**, **Business User**, or **Functional Consultant**, understanding Power BI helps you explore data and present insights instead of relying on someone else's static spreadsheet.

2. The Power BI Ecosystem

Power BI is not one single app — it is a **family of tools** that work together. The three core parts every beginner must know:

 **Power BI Desktop**

The free **Windows application** where report authors do the real work: connect to data, clean it, model it, build visuals. This is your "workshop."

Download: Microsoft Store or a direct installer. It's free.

 **Power BI Service**

The **online platform** at `app.powerbi.com`. This is where you publish, organize into workspaces, build dashboards, set refresh schedules, and share reports with others.

Needs a work/school account, or a free trial / Microsoft 365 Developer account to explore.

 **Power BI Mobile**

Cross-platform app (iOS/Android/Windows) so **consumers** can view reports on the go, in a layout optimized for small screens.

Mobile-optimized views are designed back in Power BI Desktop.

Extended ecosystem (you'll meet these as you grow)

Component	What it does
Power Query Editor	Built into Desktop (and Excel). Cleans and reshapes data before it enters your model.
Power BI Report Builder	A separate tool for pixel-perfect paginated reports (think: printable invoices, regulatory forms).
Dataflows	Reusable, cloud-based Power Query transformations that many reports can share, avoiding duplicate work.
On-premises Data Gateway	A bridge app that lets the cloud Service securely refresh data sitting inside your company's private network.
Power BI Premium / Microsoft Fabric capacity	Paid, dedicated compute/storage that unlocks larger datasets, faster refresh, and enterprise features.
Power BI Copilot	Generative-AI assistant embedded in Desktop/Service that can build visuals, summarize data and write DAX from plain English prompts.

Analogy: Desktop = kitchen where you cook. Service = restaurant where the dish is served to guests. Mobile = the take-away box guests can open anywhere.

3. The Power BI Flow — 6 Steps from Data to Decision

Every Power BI project — simple or complex — follows the same six-step flow. Memorize this order; almost everything else in the course hangs off it.



#	Step	Tool used	What actually happens
1	Connect to data	Power BI Desktop → <i>Get data</i>	Point Power BI at a source: Excel, SQL Server, SharePoint, a folder, a web API, etc.
2	Transform data	Power Query Editor	Clean column names, remove errors, split/merge columns, filter rows, change types — <i>before</i> anything is loaded into the model.
3	Model data	Power BI Desktop → Model view	Build relationships between tables, write DAX measures/calculated columns.
4	Visualize / build report	Power BI Desktop → Report view	Drag fields onto the canvas to create charts, tables, cards, slicers across one or more report pages.
5	Publish	Desktop → <i>Publish</i> button	The finished report (its semantic model + visuals) is pushed to a workspace in the Power BI Service.
6	Distribute & manage	Power BI Service	Set refresh schedules, share with colleagues, build dashboards from the report, package into an App , and secure with row-level security.

The Service can also build **high-level dashboards** that drill down into full reports, and group multiple related reports into a single easy-to-navigate **App** for end users.

4. Touring the Power BI Desktop Interface

When you open Power BI Desktop you see a **ribbon** at the top (grouped buttons, like Word/Excel), a **canvas** in the middle, and **panes** on the right. Below is a component-by-component reference matching the real Home ribbon.

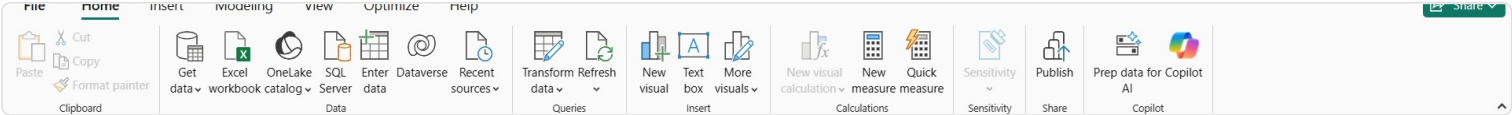


Figure 1 — The actual Home ribbon in Power BI Desktop (from your own screenshot), showing the Clipboard, Data, Queries, Insert, Calculations, Sensitivity, Share and Copilot groups described below.

4.1 The three main Views (left-hand vertical icon bar)

View icon	Purpose
Report view	Where you build pages of visuals — your main workspace.
Table/Data view	Shows raw rows of each table currently loaded into the model, like a spreadsheet grid — good for spot-checking data.
Model view	Shows all tables as boxes with lines connecting them — this is where you create/edit relationships .

4.2 Home Ribbon — Group by Group

- Clipboard** — Paste, Cut, Copy, Format painter. Same as Office apps; copies visuals/formatting between report pages.
- Data** — *Get data* (connect to a new source), *Excel workbook* (quick shortcut), *OneLake catalog / Dataverse / SQL Server* (direct connectors), *Recent sources*, *Enter data* (manually type a small table). This is Step 1 of the Flow — "Connect to data."
- Queries** — *Transform data* opens the Power Query Editor (Step 2 of the Flow). *Refresh* re-pulls the latest data from all sources into the model.
- Insert** — Add a *New visual* (blank chart placeholder), a *Text box* (titles/labels), *More visuals* (browse AppSource for custom visuals, e.g. from a marketplace).
- Calculations** — *New visual calculation*, *New measure* (write a DAX formula that calculates on the fly), *Quick measure* (pick from a template list of common calculations without writing DAX yourself).
- Sensitivity** — Apply Microsoft Purview **sensitivity labels** (e.g. "Confidential") so the report carries data-protection classification wherever it's shared or exported.
- Share** — *Publish* pushes the report to a workspace in the Power BI Service (Step 5 of the Flow).
- Copilot** — *Prep data for Copilot* and the *Copilot* pane: generative-AI helpers that can summarize your data, suggest visuals, or draft narrative insights.

4.3 The Right-Hand Panes (used constantly)

Pane	Icon looks like	What you do here
Data / Fields pane	A table/grid icon, lists tables & columns	Tick checkboxes or drag a field onto the canvas to build/extend a visual. Fields with a Σ (sigma) symbol are numeric and summarize automatically (e.g. Cost, Revenue, Units Sold in the screenshot).
Visualizations pane	Grid of small chart icons	Click an icon to change the currently selected visual's chart type (bar, line, pie, etc.). Below it: <i>Build visual</i> tab shows the field "wells" (Values, Axis, Legend...), and a second tab (paint-roller icon) opens <i>Format your visual</i> .
Filters pane	Funnel icon (left edge, labelled "Filters")	Add filters that apply to one visual, one page, or the whole report — without changing the underlying data.

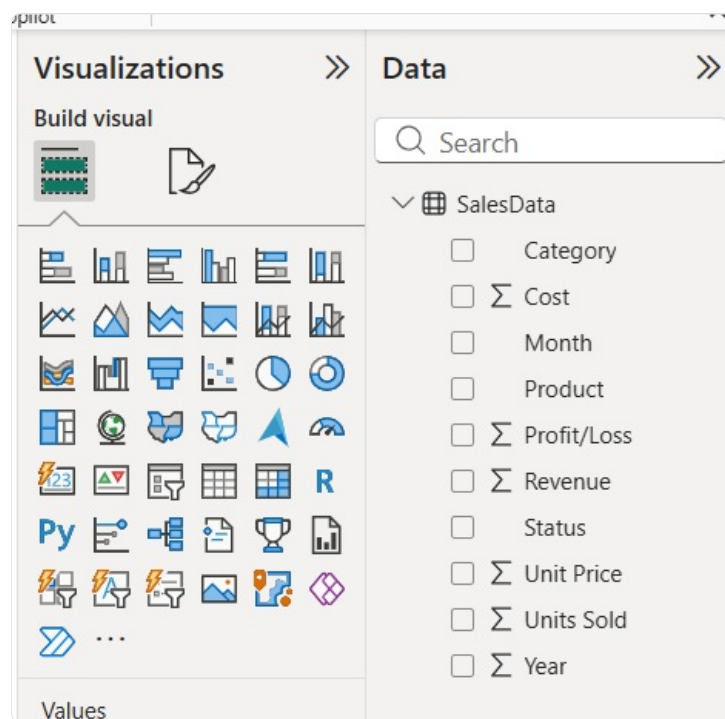


Figure 2 — The Visualizations pane, Filters pane and Data pane on the right side of Power BI Desktop, taken from your own report. Notice the Σ icons next to numeric fields (Cost, Profit/Loss, Revenue...) in the Data pane, and the Values well under "Build visual" holding Count of Product / Count of Month.

Reading the screenshot you shared: the Visualizations pane shows a *clustered column chart* ("Category" title) built from **Count of Product** and **Count of Month** on the Values well, with **Category** on the axis, plus a *Drill through* section (Cross-report: Off, Keep all filters: On) — these are report-level interactivity settings covered in Section 11.

4.4 The Rest of the Ribbon — File, Insert, Modeling, View, Optimize, Help

Your screenshot only shows the **Home** tab selected, but Power BI Desktop has several more tabs across the top: **File, Home, Insert, Modeling, View, Optimize, and Help** (plus temporary "contextual" tabs like Format, Table tools, or Column tools that appear only when you click a visual/column). Here is every one of them, button by button. (Note: exact buttons shift slightly release-to-release; this reflects the current stable ribbon layout.)

4.4.1 File tab — the "Backstage" menu

Unlike the others, File doesn't open a horizontal ribbon strip — it opens a **full-page menu** (same idea as File in Word/Excel), used for whole-file actions rather than report editing.

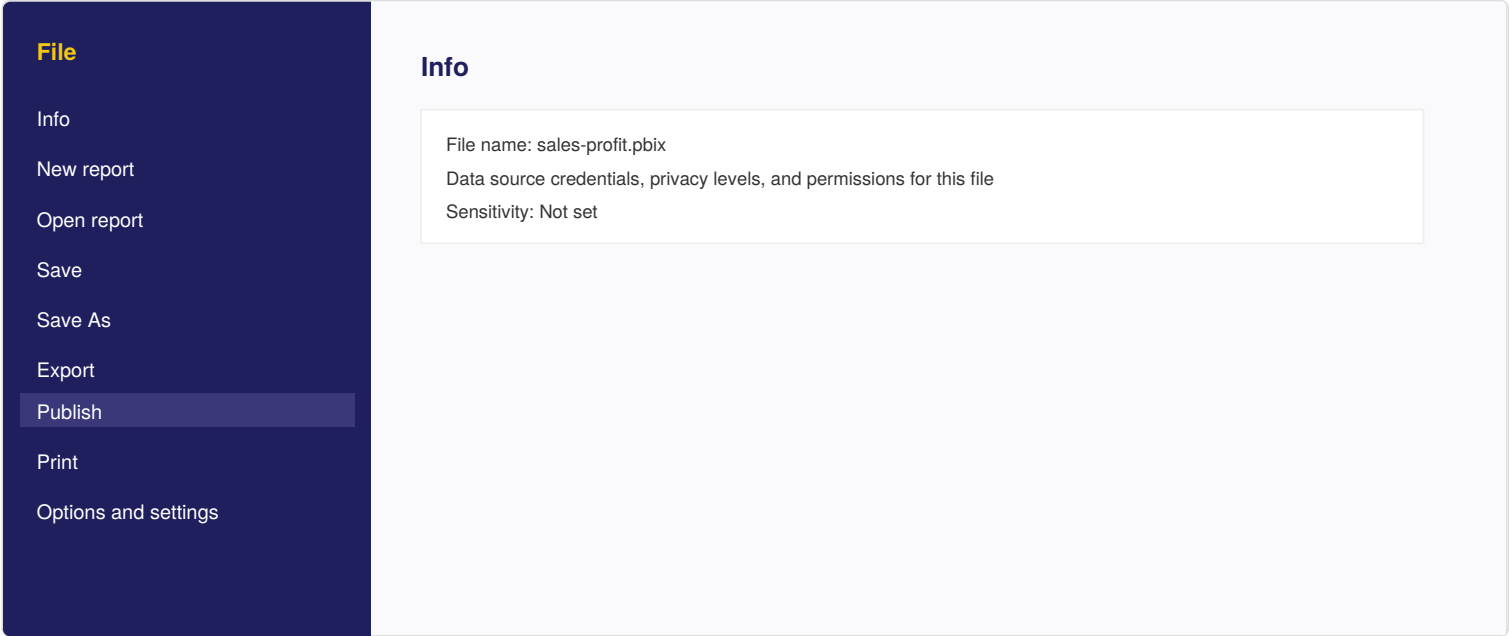


Figure 7 — The File "backstage" menu: a left-hand list of file-level actions, with details for the selected item shown on the right (here, Info).

Item	What it does
Info	Shows file name, data source credential/privacy status, and lets you review permissions before sharing the file.
New report	Starts a brand-new, blank .pbix file.
Open report	Opens an existing .pbix file from your computer or OneDrive.
Save / Save As	Saves the current file, or saves a copy under a new name/location.
Export	Export the report as a Power BI Template (.pbit), or export data/visuals to PDF or PowerPoint.
Publish	Same as the Home ribbon's Publish button — sends the report to a workspace in the Power BI Service.
Print	Prints the current report page.
Options and settings	Opens global preferences: data load options, privacy settings, preview/experimental features, regional settings, and where to manage data source credentials.

4.4.2 Insert tab — add new elements to the canvas

Everything you place onto a page *other than* dragging a field straight from the Data pane starts here.

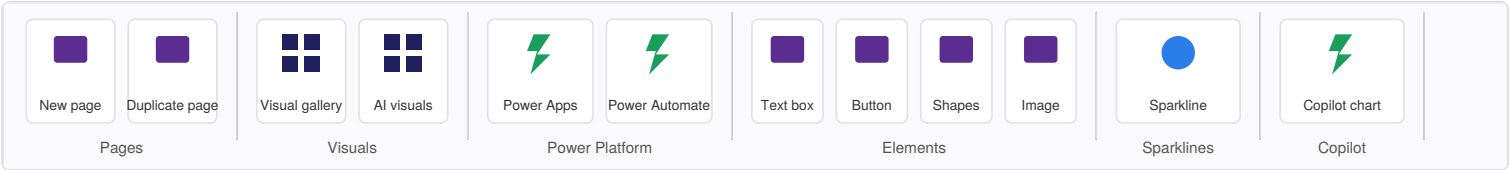


Figure 8 — The Insert tab: Pages, Visuals, Power Platform, Elements, Sparklines and Copilot groups.

Group	Buttons	Purpose
Pages	New page, Duplicate page	Add a blank report page, or clone the current page (including its visuals) as a starting point for a similar page.
Visuals	Visual gallery, AI/Copilot visuals	Same chart-type gallery as the Visualizations pane, available here too, plus AI-suggested visuals for the fields you've selected.

Power Platform	Power Apps, Power Automate	Embed a Power Apps app directly inside a report page, or add a Power Automate button visual to trigger a workflow (e.g. send an email) straight from the report.
Elements	Text box, Button, Shapes, Image	Text box — add titles/captions. Button — clickable navigation, bookmark triggers, or a "back"/"reset" action. Shapes — rectangles/lines/arrows to visually group or divide the page. Image — insert a logo or picture.
Sparklines	Sparkline	Adds a tiny trend chart inside a single table/matrix cell, showing the trend behind that one number without needing a separate visual.
Copilot	Copilot chart	Generate a new visual from a plain-English description of what you want to see.

4.4.3 Modeling tab — the data model's control room

This is where the "back-end" logic of your semantic model (Sections 5-8) is actually built and managed.

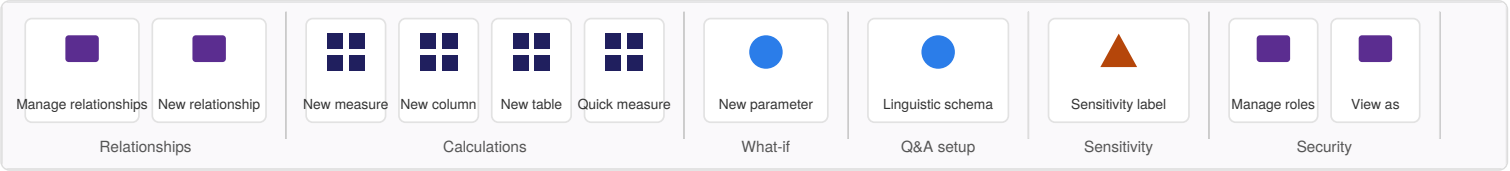


Figure 9 — The Modeling tab: Relationships, Calculations, What-if, Q&A setup, Sensitivity and Security groups.

Group	Buttons	Purpose
Relationships	Manage relationships, New relationship, Autodetect	Create/edit/delete the links between tables described in Section 7.1, or let Power BI guess relationships automatically based on matching column names.
Calculations	New measure, New column, New table, New quick measure	The main entry points for writing DAX (Section 8) — a measure (live calculation), a calculated column (row-by-row), a calculated table, or a template-based quick measure.
What-if	New parameter	Creates a What-if parameter (Section 16.2) — an adjustable, slicer-driven variable for scenario simulation.
Q&A setup	Linguistic schema, Teach Q&A	Improve how the natural-language Q&A visual (Section 10.8) interprets your field names and business terms.
Sensitivity	Sensitivity label	Same classification labels described in Section 15.2, applied here at the model/file level.
Security	Manage roles, View as	Manage roles builds Row-Level Security roles (Section 15.1). View as lets you preview the report exactly as a specific role/user would see it, before publishing.

4.4.4 View tab — control the look, feel, and workspace layout

Where Home builds content and Modeling builds logic, View controls *how the authoring environment itself looks* and how the finished report should be styled/laid out.

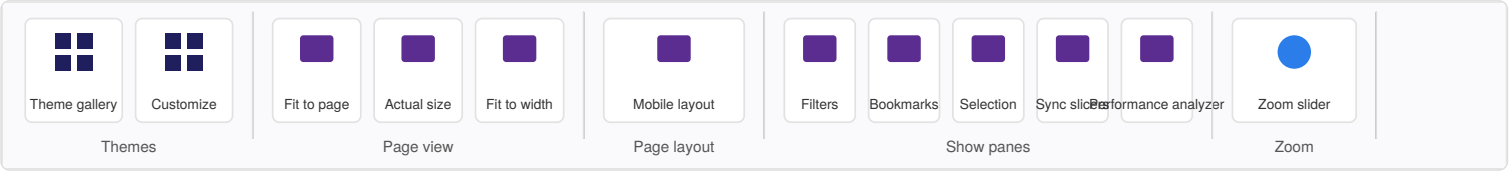


Figure 10 — The View tab: Themes, Page view, Page layout, Show panes and Zoom groups.

Group	Buttons	Purpose
Themes	Theme gallery, Customize current theme	Apply a report-wide color palette and font set in one click (Section 12), or build/import your own custom theme JSON.
Page view	Fit to page, Actual size, Fit to width	Controls how the canvas scales inside the editing window — doesn't change the report itself, just your editing view.
Page layout	Mobile layout	Switch into a phone-shaped canvas where you rearrange existing visuals into a single-column, mobile-optimized version of the page (separate from the desktop layout).
Show panes	Filters, Bookmarks, Selection, Sync slicers, Performance analyzer, and more	Tick/untick to show or hide side panes. This is the first place to look if a pane (e.g. Filters) has "disappeared."
Zoom	Zoom slider	Zoom the canvas in/out while editing (matches the zoom % shown bottom-right of the window, as in your screenshot's "82%").

Performance Analyzer (found via Show panes, or on the Optimize tab depending on version) records how long every visual takes to query and render — the main tool for diagnosing a slow report (see 4.4.5).

4.4.5 Optimize tab — make slow reports faster

Mostly relevant once your model has DirectQuery tables or has grown large enough that editing feels sluggish.

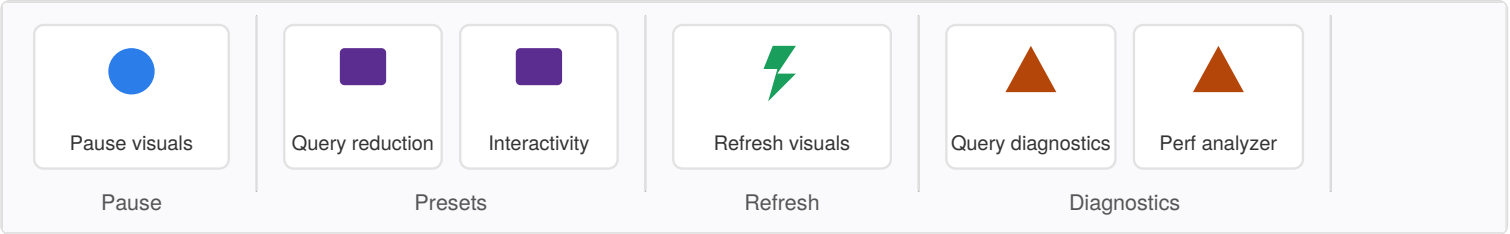


Figure 11 — The Optimize tab: Pause, Presets, Refresh, and Diagnostics groups.

Group	Buttons	Purpose
Pause	Pause visuals	Stops visuals from re-querying every time you add/move a field while you're mid-edit — add several fields, then refresh once, instead of the report re-querying after each click.
Presets	Query reduction, Interactivity	One-click bundles of settings: <i>Query reduction</i> turns off cross-highlighting/cross-filtering and adds an explicit "Apply" button to filters/slicers so changes don't fire a query on every click; <i>Interactivity</i> restores the normal instant-feedback behavior.
Refresh	Refresh visuals	Manually re-run all visual queries once, while keeping the report otherwise paused.
Diagnostics	Query diagnostics, Performance analyzer	Query diagnostics records exactly what Power Query is doing during a refresh (useful for slow data loads). Performance analyzer records how long each visual takes to query and render (useful for slow report pages).

4.4.6 Help tab — documentation, support & what's new

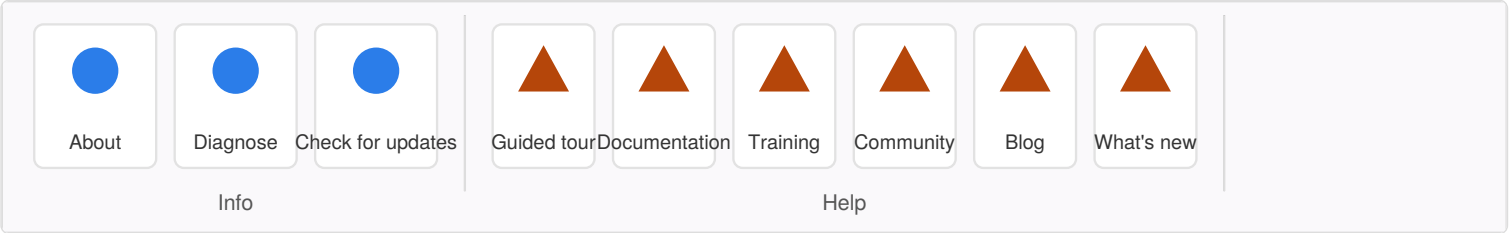


Figure 12 — The Help tab: Info and Help groups.

Group	Buttons	Purpose
Info	About, Diagnose, Check for updates	About shows your installed version number. Diagnose collects a troubleshooting log to attach to a support ticket. Check for updates looks for a newer Desktop release.
Help	Guided tour, Documentation, Training, Community, Blog, What's new	Direct links into Microsoft Learn documentation, the Power BI Community forums, the official blog (release notes), and a short interactive product tour for brand-new users.

4.4.7 Contextual tabs (only appear when something is selected)

Tab	Appears when...	Contains
Format	A visual is selected	Quick shortcuts to common formatting actions (also mirrored in the Format pane, Section 9.4).
Table tools	A table is selected in the Data pane	Rename table, mark as a Date table, manage table-level properties.
Column tools	A column is selected	Change data type/format, sort by another column, set default summarization, hide/rename.
Measure tools	A measure is selected	Format the measure's number display, change its home table, or edit its DAX directly from the ribbon's formula box.

5. Building Block 1 — The Semantic Model

Power BI has exactly **two building blocks**: the **semantic model** (data) and **visualizations** (pictures of that data). Get the model right and every visual becomes easy.

Definition: A **semantic model** (older name: "dataset") is the full package of connected data + transformations + relationships + calculations that sits behind a report. It's the single source of truth the report draws from.

Building a semantic model follows a mini-flow of its own:

1. **Connect** to one or many data sources.
2. **Clean/transform** using Power Query Editor.
3. **Add relationships** between tables and **calculations** (DAX) to extend what the raw data can answer.

Only after this is the model considered ready — *then* you build the report on top of it.

6. Power Query Editor — Getting & Transforming Data

Power Query is the "kitchen prep" stage: chop, wash, and organize data *before* it's cooked into the model. It opens from **Home → Transform data**.

Power Query Editor						
Home Transform Add Column View						
Queries [3]	OrderID	Category	Revenue	Cost	OrderDate	Applied Steps
Sales	1001	Electronics	2,500	1,800	01-05-2026	Source
Product	1002	Furniture	1,200	900	02-05-2026	Changed Type
Date	1003	Accessories	640	410	03-05-2026	Removed Columns
	1004	Electronics	3,100	2,200	05-05-2026	Filtered Rows
						Renamed Columns ★
★ = current step (each step is a recorded action)						

Figure 3 — Layout of the Power Query Editor: Queries list (left), data preview grid (center), Applied Steps recipe (right). Each step is replayed automatically on every refresh.

Common transformation	Why you'd use it
Remove/Choose columns	Drop irrelevant fields so the model stays lean and fast.
Change data type	Force a column to be Date, Whole Number, Text, etc., so calculations/sorts behave correctly.
Filter rows	Exclude test data, blanks, or out-of-scope regions/years.
Split / Merge columns	Break "First Last" into two columns, or combine "City"+"State" into one.
Group by	Pre-aggregate rows (e.g. total sales per customer) before loading.
Append queries	Stack similar tables on top of each other (e.g. Jan + Feb + Mar sales files).
Merge queries	A query-level VLOOKUP/JOIN — pull columns from one query into another based on a matching key.
Pivot / Unpivot	Reshape wide tables (one column per month) into long/tall tables (one row per month) — usually required for good modeling.
Replace values / Fill down	Fix inconsistent text entries or fill blank cells with the value above them.

Every click you make in Power Query is recorded as a step in the **Applied Steps** list (right-hand pane) — this is a repeatable, re-runnable recipe, so next time you refresh, the same cleaning happens automatically.

7. Data Modeling — Relationships & Star Schema

Once multiple tables are loaded, Power BI needs to know how they connect. This happens in **Model view**.

7.1 Relationships

Concept	Explanation
Relationship	A link between a column in Table A and a matching column in Table B (usually an ID).
Cardinality	One-to-many (1:*) is most common — e.g. one Customer has many Orders. One-to-one (1:1) and many-to-many (*:*) also exist but are less common.
Cross-filter direction	Single (default): filters flow one way, from the "one" side to the "many" side. Both : filters flow in both directions — powerful but can cause ambiguity, use carefully.
Active vs inactive relationship	Only one relationship between two tables can be "active" (solid line) by default; others are inactive (dashed) and must be invoked manually in DAX with <code>USERELATIONSHIP</code> .

7.2 Star Schema (the gold-standard model shape)

A **star schema** separates data into:

- **Fact tables** — the big transactional tables with numbers you want to measure (Sales, Orders, Revenue). Usually long and narrow.
- **Dimension tables** — descriptive lookup tables that add context (Date, Product, Customer, Region). Usually short and wide.

Fact tables connect to Dimension tables in a "star" pattern (one fact in the middle, dimensions radiating out). This shape is fast, easy to filter, and easy for report authors to understand.

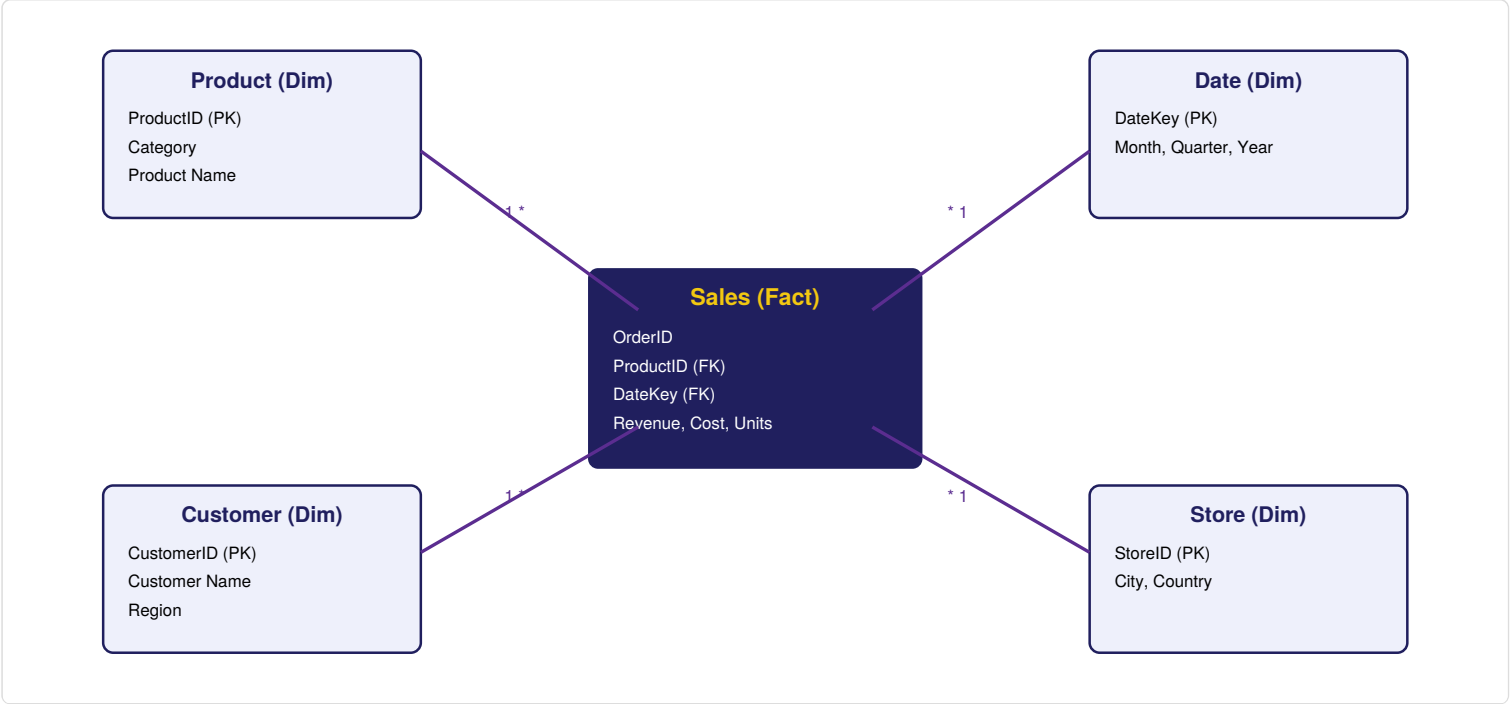


Figure 4 — A star schema as it appears in Power BI's Model view: one Sales fact table in the center, connected 1-to-many to four dimension tables (Product, Date, Customer, Store).

Advanced tip: Avoid a "snowflake" (dimensions linked to other dimensions) or one giant flat table where possible — star schema keeps DAX simpler and performance better, especially as data grows.

8. DAX — Power BI's Formula Language

DAX (Data Analysis Expressions) is the formula language used to create custom calculations in the model. If you know Excel formulas, DAX will feel familiar but works across whole tables/columns, not single cells.

8.1 The three DAX calculation types

Type	Computed when?	Stored in model?	Example use
Calculated column	Row-by-row, at refresh time	Yes, takes up memory	<code>Full Name = Customer[First] & " " & Customer[Last]</code>
Measure	On the fly, at query/visual time, based on current filters	No — calculated live	<code>Total Sales = SUM(Sales[Amount])</code>
Calculated table	At refresh time	Yes, a whole new table	A summary table or a date table generated with DAX.

Golden rule: Prefer **measures** over calculated columns whenever possible — they're lighter on memory and always reflect the current filter/slicer context (year selected, region clicked, etc.).

8.2 Must-know DAX functions

Function	What it does
<code>SUM / AVERAGE / MIN / MAX / COUNT</code>	Basic aggregations over a column.
<code>CALCULATE</code>	The most important function — re-evaluates an expression under modified filters. Powers almost every advanced measure.
<code>FILTER</code>	Returns a filtered subset of a table, usually used inside <code>CALCULATE</code> .
<code>SUMX / AVERAGEX</code>	Row-by-row calculation then aggregate — needed when a formula must multiply columns first (e.g. Qty × Price) before summing.
<code>RELATED</code>	Pulls a column value from a related table (across a relationship) into the current row.
<code>ALL / ALLEXCEPT</code>	Removes filters — useful for "% of total" calculations.
<code>DATEADD / SAMEPERIODLASTYEAR / DATESYTD</code>	Time-intelligence — compare this period vs. last, or year-to-date totals.
<code>IF / SWITCH</code>	Conditional logic.
<code>DIVIDE</code>	Safe division that avoids divide-by-zero errors.

```
Total Revenue = SUM(Sales[Revenue])
```

```
Profit Margin % =  
DIVIDE( [Total Revenue] - SUM(Sales[Cost]), [Total Revenue], 0 )
```

```
Sales LY =  
CALCULATE( [Total Revenue], SAMEPERIODLASTYEAR('Date'[Date]) )
```

Quick measures (Home ribbon → Calculations → Quick measure) let beginners pick a common calculation — running total, % of grand total, year-over-year change — from a menu, and Power BI writes the DAX for you. Great for learning DAX by example.

9. Building Block 2 — Visualizations & Reports

Once the semantic model is ready, you build a **report**: one or more pages, each page is a **canvas** holding several **visualizations** (visuals).

9.1 Key ideas

- **Low-code / drag-and-drop:** drag a field from the Data pane onto the canvas and Power BI auto-picks a sensible visual type; you can switch types anytime.
- **One idea per page:** keep each report page focused on related data so consumers can spot insights quickly — don't cram everything onto one page.
- **Interactivity:** clicking a data point in one visual automatically cross-filters/cross-highlights every other visual on the page (unless you turn this off). This is one of Power BI's biggest strengths over static slide decks.
- **Drillthrough:** consumers can right-click a data point and jump to a dedicated detail page filtered to just that item.

9.2 Reports → Dashboards

Once a report is published to the Service, you can pin individual visuals as **tiles** onto a **dashboard**. Unlike report visuals, dashboard tiles are **not interactive** — clicking one simply opens the underlying report for more detail (like a car's dashboard giving a quick glance, while the report is the full owner's manual).

9.3 Deep Dive — Every Component of the Visualizations Pane

The Visualizations pane has two tabs at the top, plus a chart-type icon grid. This is the single most-used pane in Power BI, so know it cold.

Component	What it is / does
Chart type icon grid	A grid of small icons (bar, line, pie, card, table, map, etc.). Click one to change the <i>currently selected</i> visual's type. Power BI tries to keep your field assignments when you switch — some may drop if the new chart type doesn't support them.
"..." (More visuals) icon	At the end of the grid — opens AppSource to browse/import custom, community, or organizational visuals not built into Power BI by default (e.g. Gantt chart, word cloud, radar chart).
Build visual tab (grid/bars icon)	Shows the "wells" (field drop-zones) for the selected chart — this is where you drag fields in.
Format your visual tab (paint-roller icon)	Shows all styling controls for the selected chart (see 9.4 below).
General tab (sliders icon, sometimes merged into Format)	Settings common to every visual type: title, background, border, alt text, size/position.

9.3.1 The "wells" (field drop-zones) — these change per chart type

Well name	Appears on	Purpose
Category / X-axis	Bar, column, line, most charts	The field defining categories along one axis (e.g. Product, Month).
Values / Y-axis	Almost all charts	The numeric field(s) actually being measured/plotted (e.g. Revenue). Multiple values create multiple bars/lines/series.
Legend	Bar/column/line/pie	A field that splits the data into color-coded sub-series (e.g. split Revenue bars by Region as different colors).
Small multiples	Column, line, bar, area, scatter	Splits ONE chart into a grid of many mini-versions of itself, one per category (e.g. a tiny line chart for each Region, all in a grid) — great for comparing many trends side-by-side without a crowded legend.
Tooltips	Most charts	Extra fields shown only when hovering over a data point — doesn't change the chart shape, just adds hover detail.
Secondary values / Line values	Combo charts	A second measure plotted with a different visual style (e.g. line) alongside the primary columns, often on a secondary Y-axis.
Details	Scatter/bubble chart	Defines what one dot represents (e.g. one dot per Product) so each category gets its own point.
Play axis	Scatter chart	Adds an animated time-scrubber so the scatter plot moves/evolves over a time field when played.
Rows / Columns / Values	Table, Matrix	Rows = fields going down the left; Columns = fields spread across the top (Matrix only); Values = the numbers filling the grid.
Location / Latitude / Longitude / Bubble size	Maps	Location = a place name Power BI geocodes automatically; Latitude/Longitude = precise coordinates; Bubble size = a numeric field controlling marker size.

9.4 Deep Dive — The "Format your visual" Tab

Click the paint-roller icon. Sections expand/collapse as accordions — the exact list varies slightly by chart type, but these are the ones you'll use constantly:

Section	Controls inside
Title	Text, font, size, color, alignment, background color — the chart's heading.

X-axis / Y-axis	Show/hide axis, min & max range, gridlines on/off, axis title text, number format, font.
Data colors	Pick a specific color for each category/series manually, overriding the report Theme.
Data labels	Turn on numbers directly on bars/points; control position (inside/outside), display units (thousands/millions), decimal places, font.
Legend	Show/hide, position (top/bottom/right), title text, font size.
Gridlines	The faint horizontal/vertical reference lines behind the chart — toggle and style them.
Markers	(Line/area charts) Show a dot at each data point, choose shape/size.
Plot area / Background	Fill color or image behind the chart area itself.
Border & Effects	Rounded corners, shadow, visual border thickness/color.
Header icons	Toggle small icons that appear top-right of a visual on hover: Focus mode (expand full-screen), Drillthrough/Drill-up-down arrows, "More options" (...) menu, Export data.
Alt text	A written description read by screen readers — important for accessibility.
General → Size and properties	Exact X/Y position and width/height of the visual on the canvas (for pixel-precise layouts).
General → Tooltips	Switch from the default auto-tooltip to a custom <i>Report page tooltip</i> (Section 11.4).

9.5 Deep Dive — The Data / Fields Pane

Component	Detail
Search box	Type to instantly filter the field list across all tables — fast way to find one column in a large model.
Table names	Bold headers with an expand/collapse arrow; click to reveal that table's columns and measures.
Σ (sigma) icon	Marks a numeric column/measure that will auto-summarize (Sum/Average/Count) when dragged onto a visual — e.g. Cost, Revenue, Units Sold in your screenshot.
Calendar icon	A date column — Power BI auto-builds a Year/Quarter/Month/Day hierarchy you can drill into on a visual.
fx icon	A measure (DAX calculation) rather than a raw column.
Table icon (grid)	A calculated table.
Key icon	Marks the column used as that table's unique key in a relationship.
Checkbox next to each field	Ticking it adds the field straight into the currently selected visual's default well — same effect as dragging.
Right-click menu on a field	Rename, Hide/Unhide in report view, Add to a specific well, New measure, New column, New quick measure, Sort by another column, Change summarization.

9.6 Deep Dive — The Filters Pane

Component	Detail
Filter card types	<i>Basic filtering</i> — tick values from a checklist. <i>Advanced filtering</i> — condition rules like "is greater than", "contains", "is blank". <i>Top N</i> — keep only the top/bottom N items by a measure. <i>Relative date/time</i> — dynamic windows like "last 30 days" that auto-update as time passes.
Scope headers	Fields are grouped under <i>Filters on this visual</i> , <i>Filters on this page</i> , <i>Filters on all pages</i> , and (if set up) <i>Drillthrough filters</i> — matching the three/four scopes described in Section 11.1.
Include / Exclude toggle	Flip a filter card to keep or remove the selected values.
Require single value	Forces the filter/slicer to only allow one selection at a time instead of multi-select.
Search box inside a filter card	Quickly find a value in a long list (e.g. searching a 500-customer list).
Lock icon on a filter card	Author can "lock" a filter so report viewers can see it's applied but cannot change it.
"..." on a filter card	Options to rename the filter's display label or delete the card entirely.

9.7 The Canvas & Page Controls (View ribbon)

Control	Purpose
Page size	Choose 16:9, 4:3, Letter, Tooltip (small), or a custom pixel size for the report page.

Gridlines & Snap to grid	Visual alignment guides while placing visuals; snapping keeps edges aligned automatically.
Page tabs (bottom of screen)	Add/rename/duplicate/delete/hide report pages; drag to reorder.
Page background / Canvas background	Set a color or image behind everything on the page (e.g. a subtle brand watermark).
Selection pane	Lists every object on the page like image "layers" — toggle visibility or reorder which visual sits on top when things overlap.
Sync slicers pane	Makes one slicer's selection apply across multiple pages at once, instead of resetting per page.

Trainer tip: if you only remember one thing from this deep dive — the Build tab decides *what data* appears, the Format tab decides *how it looks*, the Fields pane is *where the data lives*, and the Filters pane decides *what's excluded*. Four different jobs, four different panes.

10. Chart-by-Chart Guide — When & Why to Use Each Visual

This is the single most-asked question by beginners: "Which chart do I pick?" Use this table as your go-to reference.

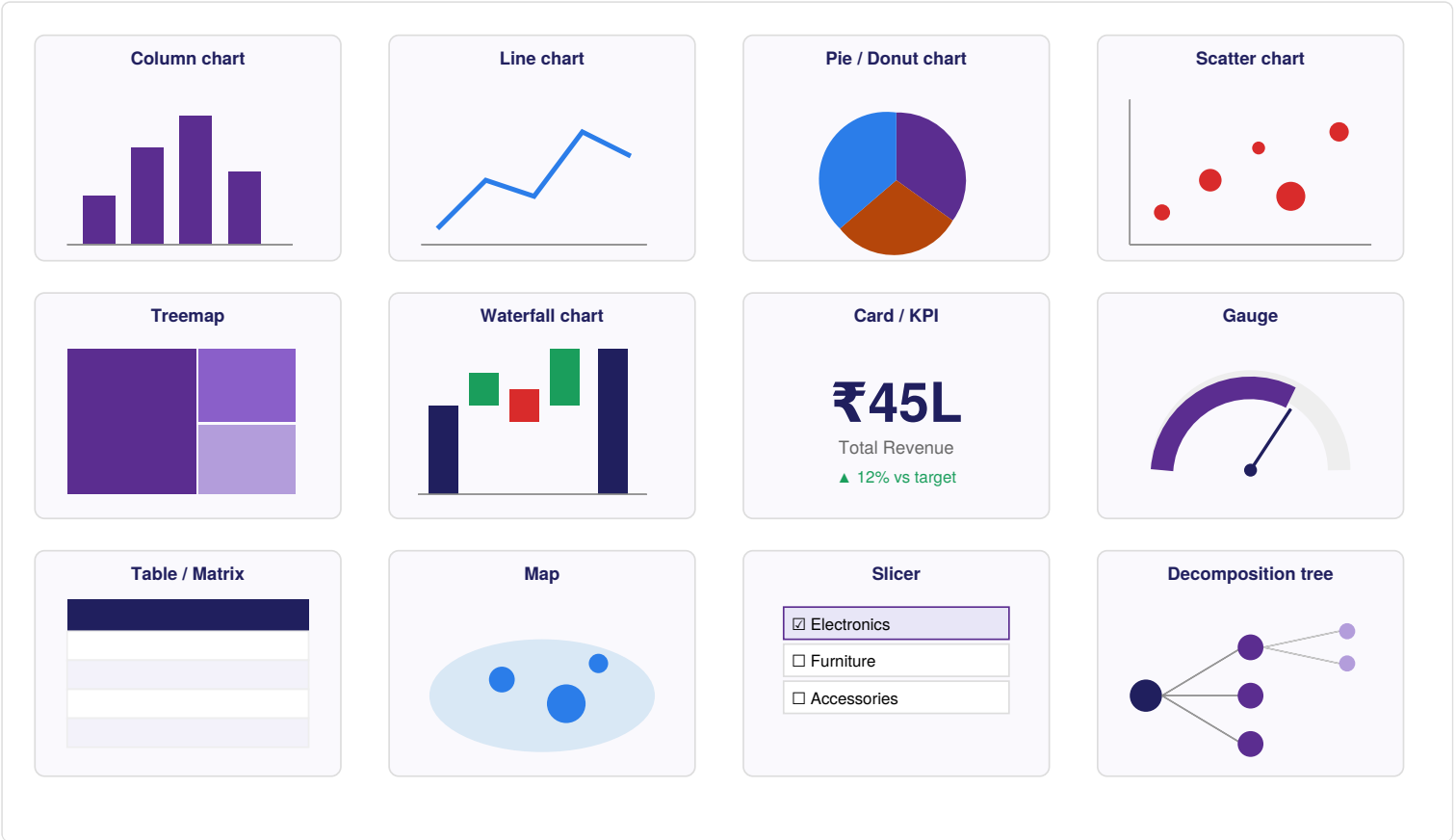


Figure 5 — Quick-glance shapes of the most common Power BI visuals: Column, Line, Pie/Donut, Scatter, Treemap, Waterfall, Card/KPI, Gauge, Table/Matrix, Map, Slicer, and Decomposition tree.

10.1 Comparing categories

Visual	Best for	Avoid when
Clustered / Stacked Column chart (vertical bars)	Comparing a numeric value across a handful of categories (e.g. Revenue by Region). Stacked also shows part-to-whole per category.	Too many categories (labels get cramped) — use a bar chart or table instead.
Clustered / Stacked Bar chart (horizontal bars)	Same as column, but better when category names are long (e.g. product names) or there are many categories to list.	Comparing trend over time — use a line chart instead.
100% Stacked Bar/Column	Showing the proportion (%) each sub-category contributes within each category.	When absolute values matter more than proportions.

10.2 Trends over time

Visual	Best for	Avoid when
Line chart	Showing a trend/change of one or more measures over a continuous axis, usually time (e.g. Monthly Revenue).	Comparing only 2-3 discrete points — a column chart reads faster.
Area chart	Same as line, but the filled area emphasizes volume/magnitude beneath the trend.	Multiple overlapping series — areas can visually hide each other; consider line chart instead.
Combo chart (line + column)	Comparing two measures with different scales or types together, e.g. Revenue (columns) vs. Profit Margin % (line).	More than 2 measures — gets cluttered.
Ribbon chart	Trend over time <i>and</i> which category ranks highest at each point — ribbons show rank changes clearly.	Too many categories — ribbons overlap and confuse.

10.3 Part-to-whole & proportion

Visual	Best for	Avoid when
Pie chart	Showing simple proportion of a whole with very few (2-5) slices.	More than ~5 categories, or comparing precise values — angles are hard to judge accurately.
Donut chart	Same use as pie, with a center hole often used to show a KPI/total.	Same limitations as pie chart.

Treemap	Part-to-whole with many categories/hierarchy levels — box size = value, great for spotting the "big few" among many small items.	Precise comparison between similarly-sized boxes is hard.
Funnel chart	Sequential stages of a process where values naturally decrease (e.g. Website Visits → Leads → Opportunities → Sales).	Data that isn't a sequential/declining process.

10.4 Relationships & distribution

Visual	Best for	Avoid when
Scatter chart	Showing the relationship/correlation between two numeric measures across many data points (e.g. Price vs. Units Sold per product).	Categorical (non-numeric) comparisons.
Bubble chart	Same as scatter, plus a third dimension shown via bubble size (e.g. adding Profit as bubble size).	Too many overlapping bubbles cause clutter.
Waterfall chart	Showing how a starting value is increased/decreased step-by-step to a final value (e.g. Budget → additions → deductions → Actual).	Non-sequential or unrelated categories.

10.5 Single-number & status visuals

Visual	Best for	Avoid when
Card	Highlighting one single important number (Total Revenue, Total Customers) prominently.	You need to show more than one metric — use a Multi-row card instead.
Multi-row card	Several individual numbers/KPIs shown as a compact list of mini-cards.	When visual comparison (bars/lines) would communicate better than raw numbers.
KPI visual	One measure tracked against a target, with a trend sparkline and up/down indicator.	No meaningful target exists to compare against.
Gauge chart	A single value against a min/max range and a target marker (speedometer style) — e.g. quota attainment.	Comparing multiple values — gauges don't compare well side-by-side.

10.6 Tabular & grid visuals

Visual	Best for	Avoid when
Table	Detailed row-level data, exact figures, exportable lists.	Audience needs a quick visual pattern — tables are read, not glanced at.
Matrix	Like a pivot table — rows × columns × values, supports drill-down through hierarchies (e.g. Year → Quarter → Month across the top, Region down the side).	Flat, non-hierarchical data — a simple table is simpler.

10.7 Geographic visuals

Visual	Best for	Avoid when
Map (bubble map)	Plotting values at specific geographic points (city, lat/long) with bubble size = value.	Precise area/region shading matters more than points.
Filled map (choropleth)	Shading whole regions/countries/states by intensity of a value.	Small/point-level location data — use bubble map instead.
Shape map / ArcGIS map	Custom boundary shapes (e.g. sales territories) or advanced GIS layers.	Simple point or region data — the standard maps are easier to set up.

10.8 AI & smart-analysis visuals

Visual	Best for	Avoid when
Key influencers	Automatically finds which factors most influence a metric going up or down (e.g. what drives high customer churn).	Very small datasets — needs enough rows to detect patterns.
Decomposition tree	Interactive drill-down/break-down of a total across multiple dimensions at once, with AI-suggested "next best" split.	Simple one-level breakdowns — a bar chart is faster to read.
Q&A visual	Consumers type a question in plain English ("total sales by region last year") and Power BI auto-generates the matching chart.	Complex custom calculations the natural-language engine can't parse.
Smart narrative	Auto-generates a text summary describing the visuals on the page (great for executive summaries).	Highly nuanced context Power BI's AI can't infer — always review the wording.

10.9 Controls & page furniture

Visual/element	Best for

Slicer	An on-canvas filter control (dropdown, list, date range slider) letting consumers filter the page/report themselves.
Button	Navigation (jump to a page), bookmarks, or triggering a "reset filters" action.
Text box	Titles, instructions, or narrative context beside charts.
Image	Logos, icons, or illustrative pictures.
Shapes	Visual grouping/dividers to organize the page layout.

Rule of thumb for choosing any chart: Ask "What question am I answering?" — *Comparison* → bar/column. *Trend* → line/area. *Proportion* → pie/donut/treemap. *Relationship* → scatter/bubble. *Single KPI* → card/gauge. *Detail* → table/matrix. *Location* → map.

11. Filters, Slicers & Interactivity

11.1 The Filters pane — three scopes

Filter level	Effect
Filters on this visual	Only affects the single selected visual.
Filters on this page	Applies to every visual on the current report page.
Filters on all pages	Applies report-wide, regardless of which page a consumer is viewing.

11.2 Slicers vs. Filters pane

A **slicer** is a visible, on-canvas control (dropdown, list, or date-range) — great for consumers who should self-serve filter without opening a side pane. The **Filters pane** is more for the report *author* to lock in a filter that isn't meant to be changed by the viewer.

11.3 Cross-filtering & cross-highlighting

Click a bar in one chart → every other visual on the page instantly updates (cross-filter) or highlights the matching portion (cross-highlight). Authors can control this per-visual via *Edit interactions* (Format ribbon).

11.4 Drillthrough & Tooltip pages

- **Drillthrough page:** a hidden detail page that a consumer reaches by right-clicking a data point → "Drill through" — automatically filtered to that item.
- **Tooltip page:** a small mini-report page that appears when hovering over a data point, instead of the default plain tooltip.

12. Formatting & the Format Pane

Every visual has a paint-roller icon opening **Format your visual**, split into two tabs:

Tab	Contains
Visual	Chart-specific settings: colors per series, axis min/max, data labels on/off, legend position, gridlines.
General	Settings shared by all visuals: title text, background, border, shadow, size & position, alt text (accessibility).

Report-wide consistency: use **Themes** (View ribbon → Themes) to apply one color palette and font set to the entire report in one click, instead of formatting every visual manually.

13. Reports vs Dashboards vs Apps vs Workspaces

Item	Lives where	Interactive?	Description
Report	Desktop & Service	Yes — full cross-filtering, drillthrough, slicers	Multi-page, deep-dive analysis built from one semantic model.
Dashboard	Service only	No (tiles link to reports)	Single page of pinned tiles for a high-level, at-a-glance summary, often combining tiles from multiple reports.
App	Service (published from a workspace)	Yes (contains real reports/dashboards)	A curated, easy-to-navigate bundle of related reports/dashboards distributed to a wider audience.
Workspace	Service	—	A container (like a folder with permissions) where a team collaborates on reports, dashboards, datasets and dataflows before optionally publishing an App from it.

14. Publishing & Sharing (Power BI Service)

Power BI Service — app.powerbi.com

Search Settings Profile

Home

Workspaces

Sales Analytics ★

Finance Team

HR Reports

Apps

Metrics

Sales Analytics workspace

+ New ↑ Publish app

All content | Reports | Dashboards | Semantic models | Dataflows

Regional Sales Report	Report	Refreshed 2h ago
Executive Dashboard	Dashboard	—
SalesData	Semantic model	Scheduled: 6am, 2pm
Sales Cleanup Flow	Dataflow	—

Roles in this workspace: Admin · Member · Contributor · Viewer

Publishing "Publish app" bundles these items into a curated App for wider distribution.

Figure 6 — Layout of a workspace inside the Power BI Service: left navigation (Workspaces, Apps, Metrics), and the workspace's content list mixing Reports, Dashboards, Semantic models and Dataflows.

14.1 Publish flow

1. In Desktop, click **Publish** (Home ribbon → Share).
2. Choose a destination **workspace** in the Service.
3. The semantic model and report are uploaded together.

14.2 Workspace roles

Role	Can do
Admin	Full control — add/remove members, delete workspace.
Member	Add/edit/publish content, share with others.
Contributor	Add/edit content, but cannot share it further.
Viewer	View-only access to content in the workspace.

14.3 Sharing methods

Method	Use when
Direct share (a link/invite to a report)	A handful of named individuals need access.
Publish an App	Distributing to a large audience with a clean, organized navigation experience.
Subscriptions	Consumers want a scheduled snapshot (e.g. emailed every Monday 9am) without opening the Service.
Embed (Power BI Embedded / "Publish to web")	Putting a report inside another application or public website.
Export (PDF/PPT/Excel)	A one-off static copy is needed, e.g. for a printed meeting.

14.4 Refresh

Datasets imported into the model don't update themselves — configure a **scheduled refresh** in the Service (up to 8x/day on Pro, more on Premium) or use **DirectQuery/real-time** connections that always query the live source.

15. Security — Row-Level Security & Sensitivity Labels

15.1 Row-Level Security (RLS)

RLS restricts which *rows* of data a user can see in a report — everyone opens the same report, but sees different data.

1. In Desktop → Modeling → **Manage roles**, create a role (e.g. "East Region") with a DAX filter: `[Region] = "East"`.
2. Publish, then in the Service, assign specific users/groups to that role under the dataset's security settings.

Dynamic RLS (advanced): instead of one static filter per role, use a mapping table plus `USERPRINCIPALNAME()` so each logged-in user automatically sees only their own rows — no need to create a role per person.

15.2 Sensitivity labels

Applied via the Home ribbon's **Sensitivity** group, these labels (Public, General, Confidential, Highly Confidential, etc., defined by your organization's Microsoft Purview policy) travel with the report/data even when exported to Excel/PDF, helping prevent accidental leaks of sensitive information.

16. Advanced Features

16.1 Bookmarks

Capture the exact state of a report page (filters applied, visuals shown/hidden, slicer selections) and return to it with one click — used to build guided "story" navigation or toggle between chart views using a button.

16.2 What-if parameters

Create an interactive slicer-driven variable (e.g. "Discount %") that feeds into a DAX measure, letting consumers simulate scenarios (e.g. "What happens to profit if I discount 10%?") live in the report.

16.3 Quick measures & visual calculations

Already introduced in Section 8 — pre-built calculation templates (running total, % of total) and, more recently, **visual calculations** that let you write a simple DAX formula directly on top of what's already rendered in a table/matrix visual.

16.4 Copilot in Power BI

Generative-AI feature (seen in the ribbon's Copilot group) that can: summarize a report in natural language, suggest and build new visuals from a text prompt, and help write/explain DAX. "*Prep data for Copilot*" adds descriptive metadata to tables/columns so Copilot's answers are more accurate.

16.5 Dataflows

Cloud-hosted, reusable Power Query transformations stored in a workspace. Multiple reports/datasets can reuse the same cleaned dataflow output instead of each duplicating the same transformation logic — improves consistency and reduces load on source systems.

16.6 Composite models & Aggregations

Composite models mix Import mode (data copied into memory — fast) and DirectQuery mode (data queried live — always current) in one model.

Aggregations pre-summarize huge fact tables so most queries hit a small, fast summary table, only reaching the full-detail table when necessary.

16.7 Incremental refresh

Instead of reloading an entire (often huge) table every refresh, incremental refresh only reloads recent partitions (e.g. "last 7 days") while historical partitions stay untouched — dramatically speeds up refresh on large datasets.

16.8 Paginated reports (Power BI Report Builder)

A separate authoring tool for **pixel-perfect, printable** reports (invoices, certificates, regulatory forms) that must render identically across page sizes — unlike interactive Power BI reports which are screen/canvas based.

16.9 On-premises Data Gateway

A lightweight app installed inside your company network that securely relays refresh requests/queries between the cloud Service and data sources that aren't publicly reachable (e.g. an internal SQL Server).

16.10 Power BI Premium / Microsoft Fabric capacity

Paid dedicated capacity (as opposed to shared Pro capacity) unlocking: larger model sizes, more frequent refreshes, paginated reports, deployment pipelines, and (via Fabric) a unified platform spanning data engineering, data science, and BI on one shared "OneLake" storage layer.

16.11 Deployment pipelines

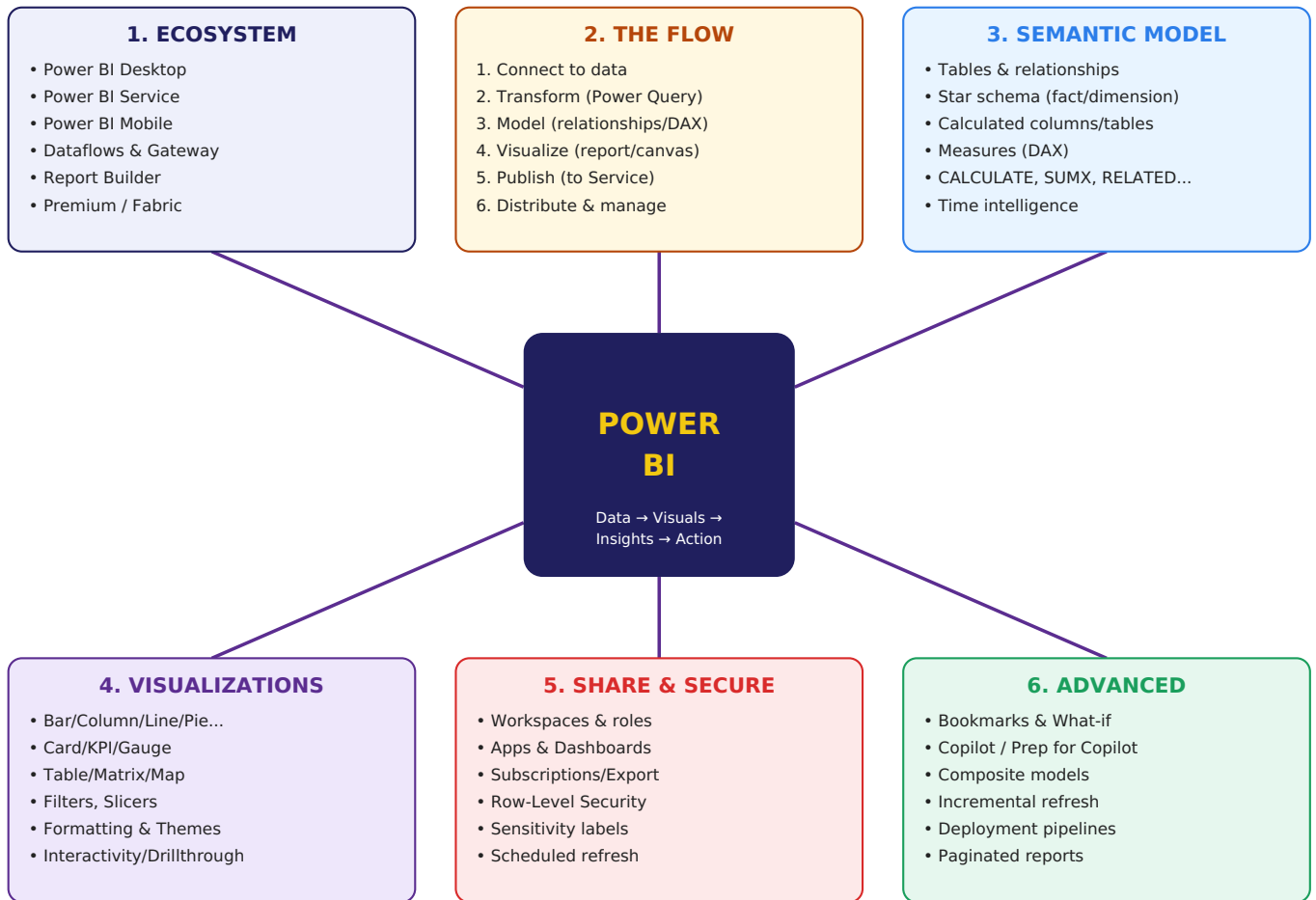
Promote content through **Development** → **Test** → **Production** workspace stages in a controlled, trackable way — similar in spirit to software release pipelines.

17. Best Practices Checklist

Area	Best practice
Modeling	Prefer a star schema; use measures over calculated columns; hide unused/technical columns from the report view.
Power Query	Filter and remove unneeded columns/rows as early as possible in the steps list — it speeds up every later step.
Report design	One clear question per page; consistent colors via Themes; limit each page to 5-8 visuals max.
Chart choice	Match the chart to the question (Section 10), not to what "looks fancy."
Performance	Use aggregations/incremental refresh on large tables; avoid excessive "Both" direction relationships and volatile DAX (e.g. NOW()) in visuals.
Security	Apply RLS and sensitivity labels before wide distribution, not after.
Governance	Use workspaces + deployment pipelines (Dev/Test/Prod) rather than editing production reports directly.
Accessibility	Add alt text to visuals, use color-blind-friendly palettes, ensure sufficient contrast.

18. Memory Map — The Whole Course on One Page

Use this mind map to revise quickly before an exam or interview — trace each branch and try to explain it out loud.



19. Glossary / Quick-Reference Cheat Sheet

Term	Meaning
Semantic model	The connected data + relationships + calculations behind a report (formerly "dataset").
Canvas	The blank page area in Report view where visuals are placed.
Visual / Visualization	A single chart, table, card, slicer, etc.
Tile	A pinned visual on a dashboard.
Workspace	A shared container in the Service for collaborating on reports/datasets.
App	A packaged, distributed bundle of reports/dashboards for end users.
Measure	A DAX calculation computed live based on filter context.
Calculated column	A DAX calculation computed row-by-row and stored in the model.
Relationship	A link between two tables via a matching column.
Cardinality	The type of relationship: one-to-many, one-to-one, many-to-many.
Star schema	Modeling pattern: fact table(s) in the center linked to dimension tables around it.
Power Query	The data-transformation engine/editor used before data loads into the model.
DAX	Data Analysis Expressions — the formula language for calculations.
Slicer	An on-canvas visual filter control.
Drillthrough	Navigating from a summary visual to a detail page filtered to the clicked item.
Bookmark	A saved snapshot of report state (filters, visibility) for one-click recall.
RLS	Row-Level Security — restricts visible rows per user/role.
Dataflow	Reusable, cloud-stored Power Query transformation shared across reports.
Gateway	Bridge app enabling the cloud Service to reach on-premises data sources.
DirectQuery	Mode where Power BI queries the source live instead of importing data.
Import mode	Mode where data is copied/compressed into Power BI's own fast in-memory engine.
Incremental refresh	Refreshing only new/changed data partitions instead of the whole table.
Paginated report	Pixel-perfect, print-ready report built in Power BI Report Builder.
Copilot	Generative-AI assistant for building visuals, summaries, and DAX from natural language.

How to keep learning: Practice the flow end-to-end on a small dataset (e.g. a sample Sales table): connect → clean in Power Query → build 2-3 relationships → write 3 measures → build one report page with a column chart, a line chart, a card, and a slicer → publish → pin a tile to a dashboard. Repeating this loop cements every concept in this document.